

1 Diagrams and summary charts for Problems 11–12

The following diagrams summarize the geometric content of the clutching construction, the unique nontrivial bundle over S^1 , and the associated sphere bundles. They are intended to support the solutions of Problems 11–12 by showing how the abstract gluing data is represented geometrically.

Vector bundles over spheres: key geometric diagrams

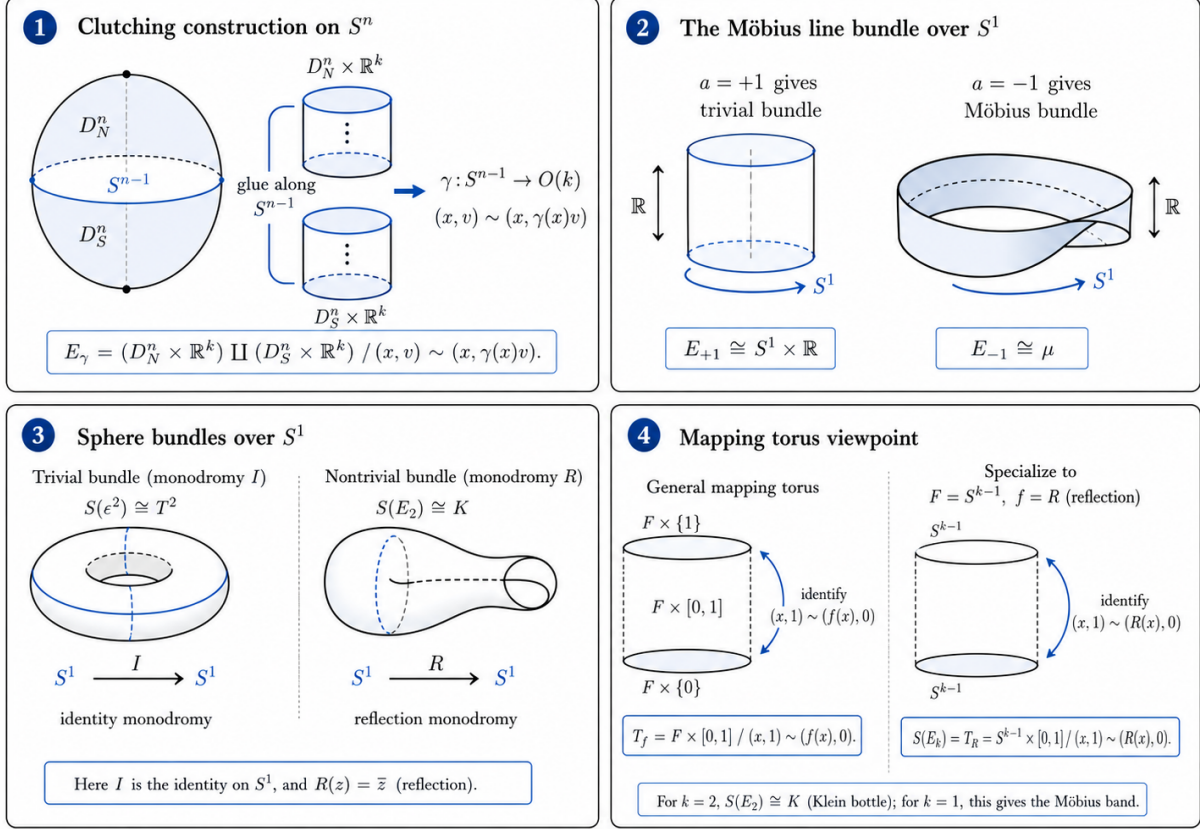
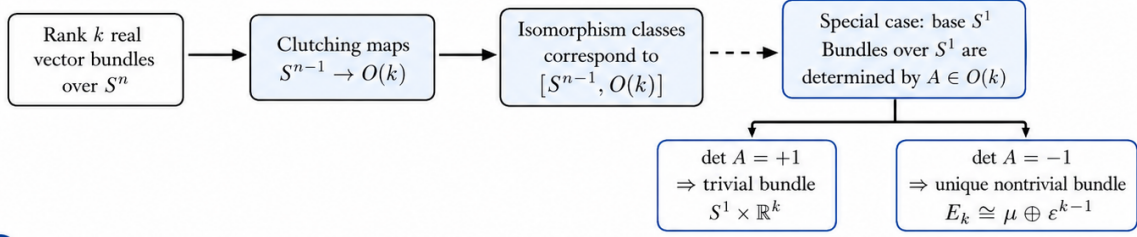


Figure 1: Key geometric diagrams for vector bundles over spheres. The first panel shows the clutching construction over S^n ; the second compares the trivial line bundle over S^1 with the Möbius line bundle; the third compares the torus and Klein bottle as sphere bundles over S^1 ; the fourth interprets these examples as mapping tori.

Classification and examples: vector bundles over spheres

1 Classification flow chart



2 Examples and comparison

Base	Rank	Clutching data	Result
S^1	1	$+1 \in O(1)$	trivial line bundle $S^1 \times \mathbb{R}$
S^1	1	$-1 \in O(1)$	Möbius line bundle μ
S^1	2	$I \in O(2)$ ($\det I = +1$)	sphere bundle = torus $T^2 = S^1 \times S^1$ (trivial)
S^1	2	$R = \text{diag}(1, -1) \in O(2)$ ($\det R = -1$)	sphere bundle = Klein bottle K
S^2	1	$S^1 \rightarrow O(1) = \{\pm 1\}$	only trivial line bundle
S^2	2	degree m map $S^1 \rightarrow SO(2) \cong S^1$	disk (or sphere) bundle with Euler number m (total space = $D(E_m)$; sphere bundle $S(E_m)$)
S^2	2	degree 2 map $S^1 \rightarrow SO(2)$	TS^2 (tangent bundle of S^2)

3 Cohomology of $S(E_k)$ for $k \geq 2$

For the unique nontrivial rank- k bundle E_k over S^1 ($E_k \cong \mu \oplus \varepsilon^{k-1}$), the cohomology of its unit sphere bundle $S(E_k)$ is	$H^q(S(E_k); \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & q = 0 \\ \mathbb{Z} & q = 1 \\ \mathbb{Z}/2 & q = k \\ 0 & \text{otherwise.} \end{cases}$	Examples: $S(E_2) \cong$ Klein bottle K. $S(E_3) =$ mapping torus of a reflection of S^2.
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Figure 2: Classification chart and examples for vector bundles over spheres. Rank k real vector bundles over S^n are described by clutching maps $S^{n-1} \rightarrow O(k)$. The special case of bundles over S^1 reduces to the determinant of the monodromy matrix $A \in O(k)$. The chart also records the cohomology of the unit sphere bundle $S(E_k)$ for the unique nontrivial bundle $E_k \rightarrow S^1$.

How to read the diagrams

In Figure ??, the bundle $E_\gamma \rightarrow S^n$ is obtained by gluing two trivial bundles over the northern and southern hemispheres. The gluing is performed along the equator S^{n-1} , and the gluing rule is encoded by the clutching map

$$\gamma : S^{n-1} \longrightarrow O(k).$$

Thus the whole bundle is determined by how the fibers are identified along the equator.

For bundles over S^1 , the monodromy matrix $A \in O(k)$ determines the bundle. If $\det A = +1$, the bundle is trivial. If $\det A = -1$, one obtains the unique nontrivial rank k bundle

$$E_k \cong \mu \oplus \varepsilon^{k-1}.$$

Its unit sphere bundle is the mapping torus of a reflection of S^{k-1} :

$$S(E_k) \cong S^{k-1} \times [0, 1] / (u, 1) \sim (Ru, 0).$$

For $k = 2$, this gives the Klein bottle:

$$S(E_2) \cong K.$$